



Important Formulas: Proportional Reasoning - 2

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Important Formulas: Proportional Reasoning - 2



Basic Proportionality

When two or more related quantities change by the same factor, they have a **proportional relationship**.

Checking Whether Two Ratios Are Proportional

Method 1: Cross-multiplication

For the ratios $a:b$ and $c:d$ to be proportional:

$$a \times d = b \times c$$

Method 2: Equal fractions

Proportional ratios can also be written as:

$$\frac{a}{b} = \frac{c}{d}$$

Equivalently, when the denominators are non-zero:

$$\frac{a}{c} = \frac{b}{d}$$

Example

Check whether 6: 3 and 4: 2 are proportional.

$$6 \times 2 = 12$$

$$3 \times 4 = 12$$

Since the two products are equal, the ratios are proportional.

Ratios with Multiple Terms

$a: b: c: d$ represents the ratio of four quantities.

If

$$a: b: c: d :: p: q: r: s$$

then the corresponding ratios are equal:

$$\frac{a}{p} = \frac{b}{q} = \frac{c}{r} = \frac{d}{s}$$

Dividing a Whole in a Given Ratio

To divide a quantity x in the ratio $a: b: c: \dots$, first find the sum of all the parts:

$$a + b + c + \dots$$

The shares are then calculated as follows:

$$\text{First part: } \frac{x \times a}{a + b + c + \dots}$$

$$\text{Second part: } \frac{x \times b}{a + b + c + \dots}$$

$$\text{Third part: } \frac{x \times c}{a + b + c + \dots}$$

Triangle Angle Division

Formula for Triangle Angles in the Ratio $a: b: c$

The sum of the three angles of a triangle is 180° . Therefore:

$$\angle A = \frac{180^\circ \times a}{a + b + c}$$

$$\angle B = \frac{180^\circ \times b}{a + b + c}$$

$$\angle C = \frac{180^\circ \times c}{a + b + c}$$

Example: Angles in the Ratio 1: 3: 5

Sum of the ratio terms:

$$1 + 3 + 5 = 9$$

Therefore:

$$\angle A = \frac{180^\circ \times 1}{9} = 20^\circ$$

$$\angle B = \frac{180^\circ \times 3}{9} = 60^\circ$$

$$\angle C = \frac{180^\circ \times 5}{9} = 100^\circ$$

Pie Chart Construction



Formula for Pie Chart Angles

The total angle of a complete circle is 360° .

For data in the ratio $a : b : c : d : \dots$, the angle corresponding to the part a is:

$$\frac{360^\circ \times a}{a + b + c + d + \dots}$$

To construct a pie chart:

1. Find the sum of all the data values.
2. Calculate the angle for each part using the formula.
3. Draw a circle and use a protractor to mark each calculated angle at the centre.

4. Label the sectors clearly.

Direct Proportion: Rule of Three

Two quantities are **directly proportional** when they both change by the same factor.

Formula

If

$$a : b :: c : d$$

then:

$$d = \frac{b \times c}{a}$$

General Relation

If x and y are directly proportional, then their corresponding ratios remain constant:

$$\frac{x_1}{y_1} = \frac{x_2}{y_2} = \frac{x_3}{y_3} = \dots = k$$

Here, k is the constant of proportionality.

Example

If 5 workers move 4500 bricks in the same amount of time, how many workers are needed to move 18000 bricks?

Since the number of workers and the number of bricks are directly proportional:

$$4500 : 18000 :: 5 : x$$

Using the rule of three:

$$x = \frac{18000 \times 5}{4500} = 20$$

20 **workers** are needed, assuming that each worker works at the same rate and the time remains the same.

Inverse Proportion

Two quantities are **inversely proportional** if, when one quantity increases by a factor of n , the other decreases by a factor of $\frac{1}{n}$.

Formula

If x and y are inversely proportional, then:

$$x_1 \times y_1 = x_2 \times y_2 = x_3 \times y_3 = \dots = k$$

Here, k is a constant.

Other equivalent forms are:

$$\frac{x_1}{x_2} = \frac{y_2}{y_1}$$

and

$$xy = k$$

Key property: If one quantity changes by a factor of n , the other changes by a factor of $\frac{1}{n}$.

Example 1: Speed and Time

For a fixed distance, speed and time are inversely proportional.

Distance = 90 km

Speed (km/h)	Time (hours)	Product
5	18	90
15	6	90
30	3	90
60	1.5	90

The relation between speed, time and distance is:

$$\text{Speed} \times \text{Time} = \text{Distance}$$

For each row in the table, the product of speed and time is 90. Thus, when speed increases, the time taken decreases so that the distance remains constant.

Example 2: Workers and Days

20 workers take 4 days to complete a task. How many days will 10 workers take to complete the same task?

For the same amount of work, the number of workers and the number of days are inversely proportional:

$$x_1 y_1 = x_2 y_2$$

$$20 \times 4 = 10 \times y_2$$

$$y_2 = \frac{20 \times 4}{10} = 8$$

8 **days** are required.

Combined Work Formula

Algebra Work Problem Formula

$$\frac{1}{t_b} = \frac{1}{t_1} + \frac{1}{t_2}$$

t_1 = time taken by first person

t_2 = time taken by second person

t_b = time taken if both do the work together

If Person A completes a work in a hours and Person B completes it in b hours:

Work done by A in 1 hour = $\frac{1}{a}$

Work done by B in 1 hour = $\frac{1}{b}$

Combined work done in 1 hour = $\frac{1}{a} + \frac{1}{b}$

Time taken to complete the work together:

$$\frac{1}{\frac{1}{a} + \frac{1}{b}}$$

Example

Ram takes 1 hour and Shyam takes 1.5 hours to cut vegetables.

Ram's work in 1 hour:

$$\frac{1}{1} = 1 \text{ unit}$$

Shyam's work in 1 hour:

$$\frac{1}{1.5} = \frac{2}{3} \text{ unit}$$

Combined work done in 1 hour:

$$1 + \frac{2}{3} = \frac{5}{3} \text{ units}$$

Time required to complete 1 unit of work:

$$1 \div \frac{5}{3} = \frac{3}{5} \text{ hours}$$

$\frac{3}{5}$ hours, or 0.6 hours, are required when they work together.

